

From Singular Learning Theory to Predictive Accuracy By Way of “Weakness as Evidence”

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Abstract

Singular Learning Theory (SLT) provides geometric invariants—most notably the local learning coefficient (LLC), also called the real log canonical threshold (RLCT), denoted λ —that govern the behavior of Bayesian evidence near singular minima. In a separate line of work, weakness theory establishes monotone relationships between a model’s “weakness” and its expected predictive accuracy in withheld-link or withheld-truth evaluation protocols. This paper unifies these two frameworks. The key observation is that weakness, when defined as the posterior-predictive mass on a truth set, can be identified with a Bayesian evidence ratio. This identification allows SLT’s asymptotic machinery to propagate into statements about “soft accuracy” (posterior-predictive mass on truth), yielding characteristic $O(\lambda/n)$ finite-sample corrections. We establish conditions under which these soft-accuracy results transfer to hard (0-1) accuracy, and argue that these conditions are typically satisfied in practical machine learning settings where models are designed to produce well-separated predictions. The framework provides a probabilistic, accuracy-facing interpretation of SLT quantities that complements standard log-loss formulations.

1 Introduction and Main Result

1.1 The problem: connecting geometry to accuracy

Singular Learning Theory (SLT) has emerged as a powerful framework for understanding Bayesian learning in models whose parameter-to-prediction map has singularities—that is, models where the Fisher information matrix degenerates at certain parameter values. Such singularities are ubiquitous in modern machine learning: neural networks with redundant hidden units, mixture models at boundaries of their parameter spaces, and many other commonly used architectures all exhibit singular behavior.

The central achievement of SLT is to characterize how these singularities affect learning. The key quantity is the *local learning coefficient* λ (also known as the real log canonical threshold or RLCT), a geometric invariant that measures the “effective dimension” of the model near a singular point. SLT shows that λ controls the asymptotic behavior of the Bayesian evidence (marginal likelihood) and, consequently, the expected generalization error—but traditionally, these results are stated in terms of log loss (KL divergence).

In many practical applications, however, the quantity of direct interest is not log loss but rather an accuracy-like metric: link prediction accuracy in knowledge graphs, retrieval hit-rate in recommendation systems, or the probability of correctly predicting withheld ground truths. The question this paper addresses is: *To what extent do SLT’s geometric invariants manifest in accuracy-based evaluation metrics?*

1.2 The challenge: from smooth to discrete

A fundamental tension underlies any attempt to connect SLT to accuracy. The quantities native to SLT—free energy, KL divergence, log loss—are smooth and differentiable. Accuracy (0-1 loss), by contrast, is a step function: it is discontinuous at decision boundaries. Improving the predicted probability of the correct class from 0.6 to 0.9 dramatically improves log loss but leaves accuracy unchanged (correct in both cases). Conversely, moving from 0.49 to 0.51 flips accuracy but barely affects log loss.

This discontinuity means that $O(\lambda/n)$ scaling, which SLT establishes for smooth quantities, cannot transfer directly to hard accuracy without additional structure. The relationship between surrogate losses (like log loss) and 0-1 loss is mediated by inequalities (e.g., Pinsker’s inequality), not equalities.

Our approach addresses this challenge by working with an intermediate quantity: *soft accuracy*, defined as the posterior-predictive mass on the truth set. This quantity is smooth (like log loss) but accuracy-adjacent (it measures probability mass on correct answers). We show that SLT’s λ/n scaling applies directly to soft accuracy. We then establish conditions—margin conditions—under which soft accuracy controls hard accuracy, and argue that these conditions are generically satisfied in practical settings.

1.3 The solution: weakness as a bridge

Our approach leverages *weakness theory*, which provides general results relating a quantity called “weakness” to expected predictive performance. The central observation is that if we define weakness as a posterior-predictive expectation—specifically, the posterior-predictive mass assigned to a “truth set” H —then weakness becomes identifiable with a ratio of Bayesian evidence integrals. Once this identification is made, SLT’s asymptotic expansions for evidence integrals yield corresponding expansions for weakness.

The logical chain proceeds as follows:

1. Define a *weakness quantity* $w(H)$ as the posterior-predictive mass on a truth set H (this is “soft accuracy”).
2. In evaluation protocols where predictions are sampled from the model distribution, establish that expected accuracy equals soft accuracy.
3. Identify weakness with a *Bayesian evidence ratio* via the posterior predictive distribution.
4. Apply *SLT evidence asymptotics* to derive the scaling behavior of weakness.
5. Conclude that soft accuracy corrections scale as λ/n (up to lower-order terms).
6. Establish conditions under which soft accuracy controls hard accuracy, and argue these hold in practice.

1.4 Statement of the main result

To orient the reader, we state our main theorem here; the concepts and notation will be developed rigorously in subsequent sections.

Theorem 1 (Main Result: From SLT to Soft Accuracy Scaling). *Fix training data D_n of size n and a Bayesian model with likelihood $p(z | w)$ and prior $\varphi(w)$. Assume the following conditions hold:*

1. (**Posterior-predictive weakness**) For a truth set $H \subseteq \mathcal{Z}$, the soft accuracy (weakness) is defined as

$$w_n(H) = \int_H p(z \mid D_n) \, dz,$$

where $p(z \mid D_n)$ is the posterior predictive distribution.

2. (**Sampling-based evaluation**) The evaluation protocol samples predictions from the model’s distribution, so that expected accuracy equals the soft accuracy $w_n(H)$.
3. (**Evidence valuation**) The posterior predictive satisfies

$$p(z \mid D_n) = \frac{\mu_n(z)}{Z_n}, \quad \text{where} \quad \mu_n(z) = \int p(z \mid w) \exp(-n\ell_n(w)) \varphi(w) \, dw.$$

4. (**SLT asymptotics**) The relevant evidence integrals admit SLT expansions with local learning coefficient λ .

Then the soft accuracy $w_n(H)$ admits a large- n expansion whose leading finite-sample correction is $O(\lambda/n)$, possibly multiplied by protocol-dependent constants. Lower-order corrections of $O(\log \log n/n)$ and $O(1/n^2)$ arise from multiplicity terms and finer geometric structure.

Remark 1 (Soft accuracy vs. hard accuracy). Theorem 1 applies directly to soft accuracy (posterior-predictive mass). For hard accuracy (0-1 loss under argmax prediction), the result extends under margin conditions that we argue are typically satisfied; see Section 7.

Remark 2 (On the rate $O(1/n)$). The $O(\lambda/n)$ rate is a “fast rate” in statistical learning parlance. Standard VC-theoretic bounds for classification error typically yield $O(1/\sqrt{n})$ rates. The faster $O(1/n)$ rate here arises from Bayesian averaging (which smooths the prediction landscape) and applies to the soft accuracy functional. The transfer to hard accuracy at this rate requires margin conditions, which we discuss in Section 7.

1.5 Scope of the results

To be precise about what this framework establishes:

Direct result: SLT’s λ controls the finite-sample behavior of posterior-predictive mass on truth sets (soft accuracy). This provides an accuracy-adjacent interpretation of λ .

Conditional extension: The transfer from soft accuracy to hard accuracy (0-1 loss) requires margin conditions—that predictions be bounded away from decision thresholds. We argue in Section 7 that these conditions are generically satisfied in well-designed machine learning systems.

Scope limitation: In adversarial or pathological settings where predictions cluster near decision boundaries, the soft-to-hard transfer may fail, and standard $O(1/\sqrt{n})$ rates would apply to hard accuracy.

2 Weakness and Expected Predictive Performance

We begin by establishing the weakness-theoretic foundation. The goal of this section is to clarify the relationship between weakness (soft accuracy) and expected accuracy under various evaluation protocols.

2.1 The withheld-link evaluation paradigm

Consider the following common evaluation scenario. We have a set of objects U and a relation $H \subseteq U \times U$ representing “true links” (or true positive pairs). Examples include:

- In a knowledge graph, U is the set of entities and H is the set of true relations.
- In a recommendation system, U is the set of users and items, and H encodes user-item preferences.
- In a social network, U is the set of users and H encodes friendships or connections.

A model is trained on a subset of the true links and evaluated on its ability to predict the withheld links.

2.2 Generalized weakness on a relation

The key quantity is the total “weight” that the model assigns to the true links.

Definition 1 (Generalized weakness on a relation). *Let $\mu : U \times U \rightarrow \mathbb{R}_{\geq 0}$ be a valuation function assigning nonnegative weights to candidate pairs. The generalized weakness (or soft accuracy) of a relation H is defined as*

$$w(H) := \sum_{(u,v) \in H} \mu(u,v).$$

The normalizing constant is

$$Z := \sum_{x,y \in U} \mu(x,y).$$

When μ is normalized so that $\sum_{x,y} \mu(x,y)/Z = 1$, the quantity $w(H)/Z$ represents the total probability mass assigned to the truth set.

Remark 3 (On the role of this definition). Defining weakness as posterior-predictive mass might seem to make subsequent results trivial (“accuracy scales like probability mass”). The substantive content comes from two sources: (1) the identification of this probability mass with evidence ratios, which connects to SLT’s geometric machinery and yields the specific λ/n scaling, and (2) the characterization of evaluation protocols under which expected hard accuracy equals soft accuracy. The definition serves as a bridge between the accuracy question we care about and the evidence integrals that SLT can analyze.

2.3 Evaluation protocols: when does soft accuracy equal expected hard accuracy?

The relationship between soft accuracy $w(H)/Z$ and expected hard accuracy depends on the evaluation protocol.

Assumption 1 (Sampling-based evaluation protocol). *A training subset $H_{\text{train}} \subset H$ is revealed to the model. At test time, predictions are generated by sampling from the model’s distribution: a pair (u,v) is predicted as a link with probability $\hat{p}(u,v) = \mu(u,v)/Z$. A test subset $H_{\text{test}} = H \setminus H_{\text{train}}$ of fixed size M is withheld, sampled uniformly among all M -subsets of H .*

Under this protocol, the expected accuracy on a single test pair $(u,v) \in H$ is exactly $\hat{p}(u,v)$, and linearity of expectation yields:

Theorem 2 (Expected accuracy under sampling-based evaluation). *Under Assumption 1, the expected accuracy on H_{test} is*

$$\mathbb{E}[\text{Accuracy}] = \frac{w(H)}{|H| \cdot Z} = \frac{1}{|H|} \sum_{(u,v) \in H} \hat{p}(u,v).$$

Proof. Because H_{test} is a uniform random size- M subset of H , and accuracy for each pair equals the predicted probability under sampling,

$$\mathbb{E}[\text{Accuracy}] = \frac{1}{M} \sum_{(u,v) \in H_{\text{test}}} \hat{p}(u,v) = \frac{1}{M} \cdot \frac{M}{|H|} \sum_{(u,v) \in H} \hat{p}(u,v) = \frac{w(H)}{|H| \cdot Z}.$$

□

Remark 4 (Contrast with argmax evaluation). Under argmax (Top-1) evaluation, the model predicts the single pair (u,v) maximizing $\mu(u,v)$. Expected accuracy then depends on whether this maximizer lies in H —a discrete event that soft accuracy does not directly control. See Section 7.

2.4 The significance of the sampling-based setting

Theorem 2 establishes that in sampling-based evaluation protocols, expected hard accuracy exactly equals soft accuracy. Many practical settings approximate this:

- Retrieval systems that return items sampled according to predicted relevance scores.
- Generative models evaluated on the quality of samples drawn from the model distribution.
- Link prediction with probabilistic thresholding.

For these settings, SLT’s control over soft accuracy translates directly to control over expected hard accuracy.

3 SLT Preliminaries: Evidence, Free Energy, and the LLC

Singular Learning Theory provides the asymptotic machinery we need to analyze how the weakness $w(H)$ behaves as a function of sample size. This section presents the essential SLT concepts.

3.1 The Bayesian setup

Let $w \in \mathcal{W} \subseteq \mathbb{R}^d$ denote the model parameters, and let $\varphi(w)$ be a prior density over $\mathcal{W}$. Given a dataset $D_n = (z_1, \dots, z_n)$, define the empirical loss as

$$\ell_n(w) := \frac{1}{n} \sum_{i=1}^n \ell(z_i; w),$$

where $\ell(z; w)$ is the negative log-likelihood for observation z under parameters w .

3.2 Evidence and free energy

The *evidence* (or marginal likelihood) is the integral of the likelihood over the prior:

Definition 2 (Evidence and free energy). *The evidence and free energy are defined by*

$$Z_n := \int_{\mathcal{W}} \exp(-n\ell_n(w)) \varphi(w) \, dw, \quad F_n := -\log Z_n.$$

One may also define local versions $Z_n(W)$ and $F_n(W)$ by restricting the integration to a neighborhood $W \subseteq \mathcal{W}$.

The evidence Z_n quantifies how well the model (integrated over all parameter values) explains the observed data. The free energy F_n is its negative logarithm.

3.3 The role of singularities

In regular statistical models—where the Fisher information matrix is positive definite everywhere—the evidence has well-known Laplace approximation asymptotics with effective dimension $d/2$. However, many models of practical interest are *singular*: the Fisher information degenerates at certain parameter values, and the standard Laplace approximation breaks down.

SLT addresses this by using tools from algebraic geometry to analyze the structure of the singularities. The key insight is that even when the parameter-to-prediction map has complex singular structure, the evidence integral still admits a controlled asymptotic expansion, but with a modified effective dimension.

3.4 The local learning coefficient

The central quantity in SLT is the *local learning coefficient* λ , also known as the *real log canonical threshold* (RLCT).

Theorem 3 (SLT local free-energy expansion (informal statement)). *Under standard SLT regularity hypotheses (analyticity of the loss function, proper prior, etc.), near a (possibly singular) minimizing set, the free energy admits an asymptotic expansion of the form*

$$F_n = n\ell_n(w^*) + \lambda \log n - (m-1) \log \log n + O_p(1),$$

where:

- w^* denotes a minimizer of the expected loss,
- $\lambda > 0$ is the local learning coefficient (LLC/RLCT),
- $m \in \mathbb{N}$ is a multiplicity term arising from the geometry of the singular set.

Exponentiating this expansion yields a corresponding form for the evidence:

$$Z_n \approx \exp(-n\ell_n(w^*)) n^{-\lambda} (\log n)^{m-1} \times (\text{bounded random factor}).$$

The LLC λ plays the role of an “effective dimension”: for regular models, $\lambda = d/2$ where d is the parameter dimension. For singular models, λ can be much smaller, reflecting the reduced effective complexity near the singularity.

4 Building the Bridge: Posterior-Predictive Weakness

We now construct the connection between weakness theory and SLT by expressing the posterior predictive as a ratio of evidence integrals.

4.1 The posterior predictive distribution

Given training data D_n , the Bayesian posterior over parameters is

$$p(w \mid D_n) = \frac{\exp(-n\ell_n(w)) \varphi(w)}{Z_n}.$$

For a candidate test item z , the *posterior predictive distribution* is

$$p(z \mid D_n) = \int p(z \mid w) p(w \mid D_n) \, dw.$$

4.2 Posterior predictive as an evidence ratio

In the log-likelihood setting where $p(z \mid w) = \exp(-\ell(z; w))$, the posterior predictive can be written as:

$$p(z \mid D_n) = \frac{1}{Z_n} \int \exp(-n\ell_n(w)) \exp(-\ell(z; w)) \varphi(w) \, dw.$$

Definition 3 (Evidence-augmented valuation). *Define the unnormalized predictive valuation by*

$$\mu_n(z) := \int \exp(-n\ell_n(w)) \exp(-\ell(z; w)) \varphi(w) \, dw.$$

This is the evidence integral for the augmented dataset $D_n \cup \{z\}$, so we write $\mu_n(z) = Z_{n+1}(z)$. The posterior predictive is the ratio

$$p(z \mid D_n) = \frac{\mu_n(z)}{Z_n} = \frac{Z_{n+1}(z)}{Z_n}.$$

This representation is crucial: it expresses the posterior predictive as a ratio of evidence integrals, both amenable to SLT analysis.

4.3 Weakness as posterior-predictive mass

Let $\mathcal{Z}$ denote the space of candidate test items, and let $H \subseteq \mathcal{Z}$ be a truth set.

Definition 4 (Posterior-predictive weakness (soft accuracy)). *The posterior-predictive weakness of a truth set H is*

$$w_n(H) := \int_H p(z \mid D_n) \, dz = \frac{1}{Z_n} \int_H \mu_n(z) \, dz.$$

With the choice $\mu = \mu_n(\cdot) = Z_{n+1}(\cdot)$, the generalized weakness becomes the posterior-predictive mass of H —exactly the soft accuracy quantity.

5 From SLT Evidence Asymptotics to Soft Accuracy Scaling

We now propagate SLT asymptotics through the posterior predictive.

5.1 The one-step expansion

Consider a single typical test item z . We have

$$\log p(z \mid D_n) = \log \mu_n(z) - \log Z_n = -F_{n+1}(z) + F_n.$$

If both $F_{n+1}(z)$ and F_n admit SLT expansions with the same dominant stratum and the same LLC λ , we can compute:

$$\begin{aligned} F_{n+1} - F_n &\approx \ell(w^*) + \lambda(\log(n+1) - \log n) + (\text{lower order}) \\ &= \ell(w^*) + \lambda \log\left(1 + \frac{1}{n}\right) + \dots \\ &= \ell(w^*) + \frac{\lambda}{n} + O\left(\frac{1}{n^2}\right). \end{aligned}$$

Therefore,

$$\log p(z \mid D_n) \approx -\ell(w^*) - \frac{\lambda}{n} + O\left(\frac{1}{n^2}\right).$$

5.2 Soft accuracy scaling

Since $w_n(H) = \int_H p(z \mid D_n) \, dz$, if $p(z \mid D_n)$ has a multiplicative perturbation of the form

$$p(z \mid D_n) = p_\infty(z) \left(1 - c \frac{\lambda}{n} + o\left(\frac{1}{n}\right)\right)$$

uniformly over $z \in H$, then

$$w_n(H) = w_\infty(H) \left(1 - c \frac{\lambda}{n} + o\left(\frac{1}{n}\right)\right).$$

This establishes the $O(\lambda/n)$ scaling for soft accuracy.

6 The Complete Argument

Proof of Theorem 1. Step 1: By Theorem 2, under sampling-based evaluation, expected hard accuracy equals soft accuracy $w_n(H)/(|H| \cdot Z_n)$.

Step 2: By the evidence-augmented valuation construction, $w_n(H) = Z_n^{-1} \int_H \mu_n(z) \, dz$.

Step 3: Each $\mu_n(z) = Z_{n+1}(z)$ and Z_n admit SLT expansions. The ratio satisfies

$$\log p(z \mid D_n) = -\ell(w^*) - \frac{\lambda}{n} + O\left(\frac{1}{n^2}\right) + O\left(\frac{\log \log n}{n}\right).$$

Step 4: Integrating over H yields

$$w_n(H) = w_\infty(H) \left(1 - c \frac{\lambda}{n} + \text{lower order}\right).$$

Step 5: Under sampling-based evaluation, this directly yields the $O(\lambda/n)$ correction for expected accuracy. $\square$

7 Extension to Hard Accuracy

Theorem 1 applies to soft accuracy and to expected hard accuracy under sampling-based evaluation. This section addresses hard accuracy under argmax (Top- k) evaluation.

7.1 The decision boundary problem

Under argmax evaluation, the model predicts the item(s) with highest $\mu(z)$. Accuracy is 1 if the argmax lies in H , and 0 otherwise. This is a discontinuous function of the model parameters: small changes in μ that shift the argmax across the boundary of H cause discontinuous jumps in accuracy.

Soft accuracy $w_n(H)$ does not directly control whether the argmax lies in H . A model with $w_n(H) = 0.9$ might have its argmax outside H (if the remaining 0.1 mass is concentrated on a single high-scoring false positive).

7.2 Margin conditions

The transfer from soft to hard accuracy requires a *margin condition*: the posterior predictive must be bounded away from 0.5 (or more generally, from the decision threshold) on the truth set.

Definition 5 (Margin condition). *A model satisfies a margin condition with parameter $\gamma > 0$ if, for all $z \in H$,*

$$p(z \mid D_n) \geq \frac{1}{2} + \gamma.$$

Under such a condition, improvements in soft accuracy translate to improvements in hard accuracy, and the $O(\lambda/n)$ rate can transfer. This is analogous to the Tsybakov margin condition in statistical learning theory, which enables fast $O(1/n)$ rates for classification.

Proposition 1 (Soft-to-hard transfer under margin conditions). *If the margin condition holds uniformly over H with parameter $\gamma > c\lambda/n$ for all sufficiently large n , then the $O(\lambda/n)$ scaling of soft accuracy implies $O(\lambda/n)$ scaling of hard accuracy.*

7.3 Why margin conditions typically hold in practice

While margin conditions are not guaranteed by SLT alone, there are several reasons to expect they will be satisfied in practical machine learning settings.

1. Posterior concentration in well-specified models. In Bayesian inference, when the model is well-specified (i.e., the true data-generating process lies within the model class), the posterior concentrates around the true parameters as $n \rightarrow \infty$. This concentration implies that posterior-predictive probabilities for true items converge to their true probabilities, which are bounded away from decision thresholds by definition. More precisely, if the true probability of item $z \in H$ is $p^*(z) > 1/2 + \delta$ for some $\delta > 0$, then posterior concentration ensures $p(z \mid D_n) > 1/2 + \delta/2$ for sufficiently large n .

2. The structure of link prediction and retrieval tasks. In link prediction, the truth set H consists of actual relationships (e.g., real friendships, verified facts, observed interactions), while the complement consists of non-relationships. Models are specifically designed to exploit features that distinguish true links from false ones—co-occurrence patterns, structural properties, semantic similarity, etc. By construction, these features create a substantial gap between scores for true and

false items. A model that did not produce such separation would be useless for the task and would not be deployed.

Quantitatively, if a link prediction model achieves (say) 80% ranking accuracy, then for a typical true link $(u, v) \in H$, the model scores (u, v) higher than 80% of false candidates. This ranking separation implies a corresponding gap in probability space after normalization.

3. Bayesian averaging smooths borderline cases. The posterior predictive $p(z \mid D_n) = \int p(z \mid w)p(w \mid D_n)dw$ averages predictions over the posterior distribution of parameters. This averaging tends to smooth out extreme predictions: if some parameter values yield $p(z \mid w) \approx 0.5$ while others yield confident predictions, the average will typically be pulled toward the confident majority (assuming the posterior concentrates on good parameter regions). This smoothing effect reduces the fraction of test items near the decision boundary.

4. Hard cases are typically rare. Even when some items in H violate the margin condition, these “hard cases” are typically a small fraction of the test set. Let $H_{\text{hard}} \subseteq H$ be the set of items with $|p(z \mid D_n) - 1/2| < \gamma$, and let $H_{\text{easy}} = H \setminus H_{\text{hard}}$. If $|H_{\text{hard}}|/|H| = o(1)$ as $n \rightarrow \infty$, then the overall accuracy is dominated by H_{easy} , on which the $O(\lambda/n)$ scaling applies.

Empirically, calibration studies of modern machine learning models consistently find that confident predictions (far from 0.5) substantially outnumber uncertain predictions (near 0.5) on typical test sets. This is not coincidental: it reflects the fact that most test examples are “typical” instances well-represented in training data, while only a minority are genuinely ambiguous.

5. The margin itself improves with n . As sample size increases, the posterior concentrates, and margin parameters γ typically grow (or at least remain bounded away from zero). The condition $\gamma > c\lambda/n$ in Proposition 1 becomes easier to satisfy as $n \rightarrow \infty$, since the right-hand side shrinks. Thus, even if margins are small for finite n , they eventually dominate the λ/n correction term.

Remark 5 (When margins fail). Margin conditions can fail in adversarial settings: if an adversary constructs test items specifically chosen to lie near decision boundaries, or if the truth set H is defined to include only ambiguous cases. They can also fail in severely misspecified models where the true data distribution places substantial mass near decision boundaries. These pathological cases are important for worst-case analysis but are not representative of typical machine learning deployments.

7.4 A refined statement

Combining Theorem 1 with the margin analysis yields:

Corollary 1 (Hard accuracy scaling under typical conditions). *Under the hypotheses of Theorem 1, if additionally:*

- (i) *The model is approximately well-specified, or*
- (ii) *The task structure ensures separation between true and false items, or*
- (iii) *The fraction of test items near decision boundaries is $o(1)$,*

then expected hard accuracy under argmax evaluation admits $O(\lambda/n)$ finite-sample corrections on the “easy” subset of the test set, and overall accuracy corrections are $O(\lambda/n) + o(1)$.

8 Discussion

1. The role of Bayesian averaging. The $O(1/n)$ rate—faster than the typical $O(1/\sqrt{n})$ of VC theory—arises from Bayesian averaging. The posterior predictive integrates over all parameter values, smoothing the prediction landscape. This smoothing is what allows SLT’s analysis (which targets smooth functionals) to apply. Empirical risk minimization with a fixed parameter estimate would not enjoy the same smoothness, and standard $O(1/\sqrt{n})$ rates would apply.

2. Soft accuracy as the natural SLT target. Soft accuracy (posterior-predictive mass) is the natural accuracy-like quantity for SLT analysis. It is smooth, it equals expected hard accuracy under sampling-based evaluation, and it admits direct SLT asymptotics. Hard accuracy under argmax evaluation is a derived quantity that requires additional (margin) conditions—but as argued above, these conditions are typically satisfied.

3. Multiple strata and phase transitions. When multiple singular strata compete, SLT predicts switching behavior driven by the tradeoff between the leading $n\ell$ term and the subleading $\lambda \log n$ term. In such cases, soft accuracy can exhibit stage-like transitions rather than smooth $1/n$ decay. The $O(\lambda/n)$ scaling applies within each regime but may not capture inter-regime transitions.

4. Connection to scaling laws. Modern deep learning research emphasizes scaling laws where loss decreases as $N^{-\alpha}$ with data or model size. The λ/n correction is consistent with this picture, with the added nuance that λ is not constant—it depends on which singular stratum the model occupies. A model finding a flatter minimum (lower λ) will have a smaller coefficient in front of $1/n$, generalizing faster for the same data.

5. Weakness as an Occam’s Razor formalization. Identifying weakness with evidence ratios connects to algorithmic information theory: the model seeks the lowest-complexity (λ) explanation of the data. Lower λ implies both better compression (in the information-theoretic sense) and higher soft accuracy on withheld truths.

9 Conclusion

By defining weakness as the posterior-predictive mass on a truth set H and identifying this with a Bayesian evidence ratio, we have shown that SLT’s $\lambda \log n$ term propagates into soft accuracy, yielding characteristic $O(\lambda/n)$ finite-sample corrections. The framework establishes a clear chain from SLT’s geometric invariants to accuracy-like quantities:

- **Direct:** λ controls soft accuracy (posterior-predictive mass on truth).
- **Under sampling evaluation:** Soft accuracy equals expected hard accuracy.
- **Under argmax evaluation:** Soft accuracy controls hard accuracy when margin conditions hold—and we have argued these conditions are typically satisfied in practical settings where models are designed to produce well-separated predictions.

This provides an accuracy-facing interpretation of SLT: the local learning coefficient λ governs not only log-loss generalization but also, under typical conditions, the rate at which predictive accuracy improves with sample size.

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